

Modelling Volatility II: The GARCH Model

Week 2, Day 2 – Advanced Financial Econometrics

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Reference: Tsay, *Analysis of Financial Time Series*, Ch. 3 (Bollerslev, 1986).

All empirical results use the course **data/** folder: S&P 500, Apple, Microsoft daily returns, 2000–2026.

Where we were, and where we go today

- **Yesterday.** ARCH made the *variance* time-varying: $\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i a_{t-i}^2$. It captured clustering and made risk *forecastable*.
- **But it paid a price.** On daily equities the volatility memory is *long*: ARCH(1) $\rightarrow$ (2) $\rightarrow$ (5) kept improving but never cleaned the residuals, and each extra day of memory cost *another* parameter.

Today's question

Can we match that long memory *cheaply* – with a handful of parameters instead of dozens? The answer is Bollerslev's **GARCH**: let the variance remember *its own past*, not just past shocks.

- Build **GARCH**(1,1) step by step, and see why two extra letters buy an *infinite* ARCH.
- Estimate, diagnose and **forecast** it on real S&P 500, Apple and Microsoft data.

Roadmap

- ➊ Why ARCH is not enough
- ➋ The GARCH model, step by step
- ➌ Why GARCH works: properties
- ➍ Building a GARCH model
- ➎ Student- t innovations
- ➏ Forecasting volatility
- ➐ Does it travel? Cross-asset evidence
- ➑ Bringing it together: a worked exercise

Why ARCH is not enough

The one weakness of ARCH: it is a spendthrift with lags

Recall yesterday's order study (daily S&P 500)

Order	Params	Persistence $\sum_i \hat{\alpha}_i$	Leftover kurtosis	Ljung-Box $\hat{\varepsilon}_t^2$
ARCH(1)	2	0.39	10.3	≈ 0
ARCH(2)	3	0.61	6.5	≈ 0
ARCH(5)	6	0.82	4.7	9×10^{-10}

- Each extra day of memory demands *another* α_i – and even five lags do not fully clean the residuals.
- **The pattern in the $\hat{\alpha}_i$ is the clue:** they decay *smoothly* (roughly geometrically) across lags. Estimating a dozen freely-moving, near-collinear weights is wasteful and **fragile**.

We want the same slowly-decaying memory, but described by one number, not a dozen.

The idea that saves parameters: recursion

A geometric memory can be written recursively

A weighted sum with geometrically-decaying weights,

$$\sigma_t^2 = c + \alpha(a_{t-1}^2 + \beta a_{t-2}^2 + \beta^2 a_{t-3}^2 + \dots),$$

obeys a **one-line recursion** – yesterday's whole sum is hidden inside yesterday's variance:

$$\sigma_t^2 = \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2.$$

- The infinite tail of past shocks is carried, for free, by the single term $\beta \sigma_{t-1}^2$.
- **Three numbers** (ω, α, β) now encode an *unbounded* memory.
- That recursion *is* the GARCH model. The rest of today unpacks it.

The GARCH model, step by step

Bollerslev (1986): let variance feed on its own past

The GARCH(1,1) model – Generalised ARCH

$$\begin{aligned}a_t &= \sigma_t \varepsilon_t, & \varepsilon_t &\stackrel{iid}{\sim} (0, 1), \\ \sigma_t^2 &= \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2.\end{aligned}$$

- Same first line as ARCH: a shock is volatility $\times$ pure noise. The novelty is the $\beta \sigma_{t-1}^2$ term.
- Read it as **three forces on today's variance**:
 - ω – a **baseline** (the floor variance in perfect calm);
 - αa_{t-1}^2 – the **news / reaction**: how hard yesterday's *shock* moves it;
 - $\beta \sigma_{t-1}^2$ – the **memory / inertia**: how much of yesterday's *variance* persists.
- “(1,1)” = one lag of a^2 and one lag of σ^2 . It is, in practice, almost always enough.

Reading the variance equation carefully

$$\sigma_t^2 = \omega + \underbrace{\alpha a_{t-1}^2}_{\text{news}} + \underbrace{\beta \sigma_{t-1}^2}_{\text{memory}}$$

- Like ARCH, σ_t^2 is **known at time $t-1$** : once a_{t-1} and σ_{t-1}^2 are in hand, today's variance is a fixed number – only the *sign* of a_t stays random.
- **α vs β do different jobs.** α sets how sharply volatility *jumps* on fresh news; β sets how slowly it *fades* afterwards. High β = long, smooth memory.
- For daily equities the estimates are almost always **small α , large β** ($\alpha \approx 0.1$, $\beta \approx 0.85$): volatility reacts modestly to any one day but forgets slowly.

Why this beats a high-order ARCH

A big β reproduces the effect of *many* ARCH lags with a *single* parameter – the parsimony ARCH lacked. We prove this on the next slide.

Constraints – what (ω, α, β) are allowed to be

1. A variance can never be negative

$\sigma_t^2 \geq 0$ for every history $\implies \omega > 0, \alpha \geq 0, \beta \geq 0$.

2. The variance must not explode – (weak) stationarity

A finite, constant long-run variance requires $\alpha + \beta < 1$. The sum $\alpha + \beta$ is the **persistence**: how much of a variance shock survives to the next day. If $\alpha + \beta \geq 1$, shocks never fully die – the process is non-stationary (that limit is *IGARCH*).

3. Finite fourth moment (for a finite kurtosis)

Gaussian ε_t : additionally $(\alpha + \beta)^2 + 2\alpha^2 < 1$.

Each is again just “a variance / a kurtosis must exist.”

Why GARCH works: properties

Property 1 – GARCH(1, 1) is an ARCH(∞)

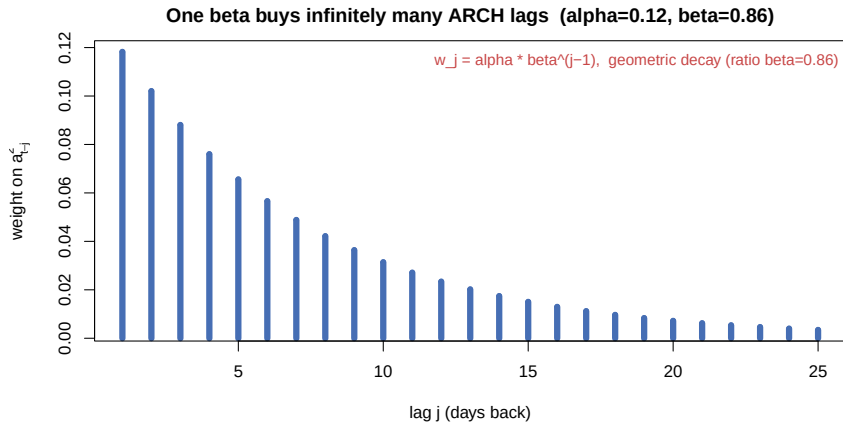
Unfold the recursion

Substitute $\sigma_{t-1}^2 = \omega + \alpha a_{t-2}^2 + \beta \sigma_{t-2}^2$ into σ_t^2 , again and again:

$$\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha \sum_{j=1}^{\infty} \beta^{j-1} a_{t-j}^2.$$

- So GARCH(1, 1) **is** an ARCH with *infinitely many* lags, whose weights $\alpha\beta^{j-1}$ decay **geometrically** at rate β .
- **One β buys the whole infinite tail.** That is the parsimony we were missing: 3 parameters, unbounded memory.
- Recent shocks matter most; older ones fade but never vanish abruptly – exactly the smooth decay we saw in the ARCH(5) weights.

The geometric weights, drawn



Each bar is the weight $\alpha\beta^{j-1}$ that GARCH(1,1) (fitted S&P 500) places on the squared shock j days ago. With $\beta = 0.86$, the weight at lag j is 86% of the weight at lag $j - 1$: a shock still carries measurable weight *weeks* later. An ARCH would need one free parameter per bar; GARCH needs one β for all of them.

Property 2 – the unconditional variance

Take expectations of the recursion

With $\mathbb{E}[a_t^2] = \mathbb{E}[\sigma_t^2] \equiv \sigma^2$ under stationarity,

$$\sigma^2 = \omega + \alpha \sigma^2 + \beta \sigma^2 \implies \boxed{\sigma^2 = \frac{\omega}{1 - \alpha - \beta}} \quad (\alpha + \beta < 1).$$

- The closer $\alpha + \beta$ is to 1, the *larger* and more sensitive this long-run level – and the slower the reversion to it.
- **Two horizons again:** the variance wobbles day to day but is tethered to a fixed long-run anchor $\omega/(1 - \alpha - \beta)$. Volatility **mean-reverts**.

Property 3 – persistence and the volatility half-life

How long does a volatility shock last?

The h -step forecast decays toward the long-run level as $(\alpha + \beta)^h$. The **half-life** – days until half the deviation is gone – is

$$H = \frac{\ln 0.5}{\ln(\alpha + \beta)}.$$

- **S&P 500:** $\alpha + \beta = 0.981 \Rightarrow H \approx 37$ trading days ($\sim$ seven weeks). A spike in volatility is *half*-forgotten only after seven weeks – genuinely long memory.
- This single number, $\alpha + \beta$, is what an entire ARCH(q) struggled to reach with many lags.

Property 4 – the squares follow an ARMA(1, 1)

Same trick as ARCH: let $\eta_t = a_t^2 - \sigma_t^2$ (a mean-zero surprise)

Substituting $\sigma_t^2 = a_t^2 - \eta_t$ into the GARCH recursion gives

$$a_t^2 = \omega + (\alpha + \beta) a_{t-1}^2 + \eta_t - \beta \eta_{t-1}.$$

- So a_t^2 is an **ARMA(1, 1)** with AR coefficient $\alpha + \beta$ – persistent but stationary when $\alpha + \beta < 1$.
- Its autocorrelations decay slowly at rate $\alpha + \beta$: exactly the long, gently-fading ACF of squared returns we could never fit with a low-order ARCH.
- **Punchline:** a_t stays white noise, yet a_t^2 has rich ARMA structure – clustering and unpredictable direction, together, from three parameters.

Building a GARCH model

The workflow is unchanged – only the variance equation grew

- ➊ **Model the mean.** Fit an ARMA to r_t (usually just a constant); take residuals a_t .
- ➋ **Test for ARCH effects.** Ljung–Box on a_t^2 / Engle's LM (as yesterday).
- ➌ **Estimate ω, α, β by maximum likelihood** – the same Gaussian log-likelihood, now with the GARCH recursion for σ_t^2 .
- ➍ **Check the fit.** Standardised residuals $\hat{\varepsilon}_t = a_t / \hat{\sigma}_t$ should be i.i.d. with *no* remaining ARCH.
- ➎ **Compare models.** AIC/BIC and the $\hat{\varepsilon}_t^2$ diagnostics decide ARCH vs GARCH.

The one new subtlety in estimation

σ_t^2 now depends on σ_{t-1}^2 , so the likelihood is built by **recursion**: pick a starting σ_0^2 (e.g. the sample variance), then roll $\sigma_t^2 = \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2$ forward and sum the log-density. Maximise over (ω, α, β) .

GARCH(1, 1) fitted to daily S&P 500 (MLE)

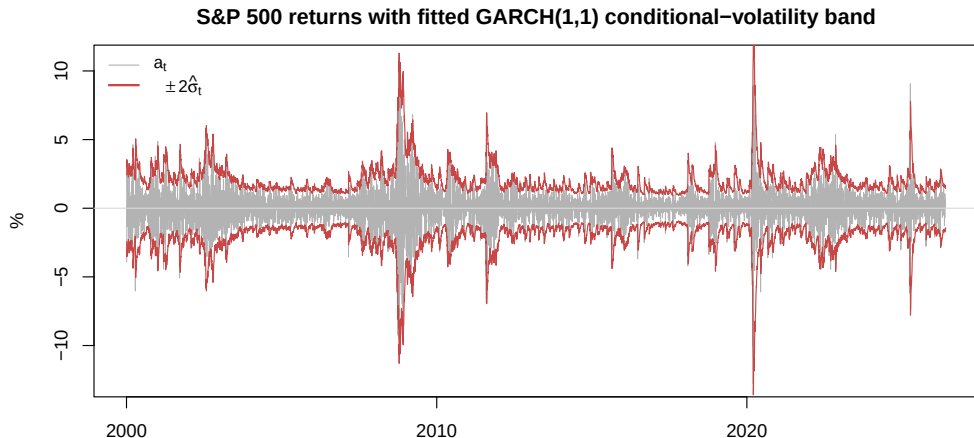
The fit

$$\hat{\sigma}_t^2 = \underbrace{0.0245}_{(\hat{\omega})} + \underbrace{0.118}_{(\hat{\alpha})} a_{t-1}^2 + \underbrace{0.863}_{(\hat{\beta})} \sigma_{t-1}^2.$$

$$\hat{\alpha} + \hat{\beta} = \mathbf{0.981} \text{ (persistence),} \quad \frac{\hat{\omega}}{1 - \hat{\alpha} - \hat{\beta}} = 1.30 \Rightarrow \hat{\sigma} = \mathbf{1.14\% \text{ per day.}}$$

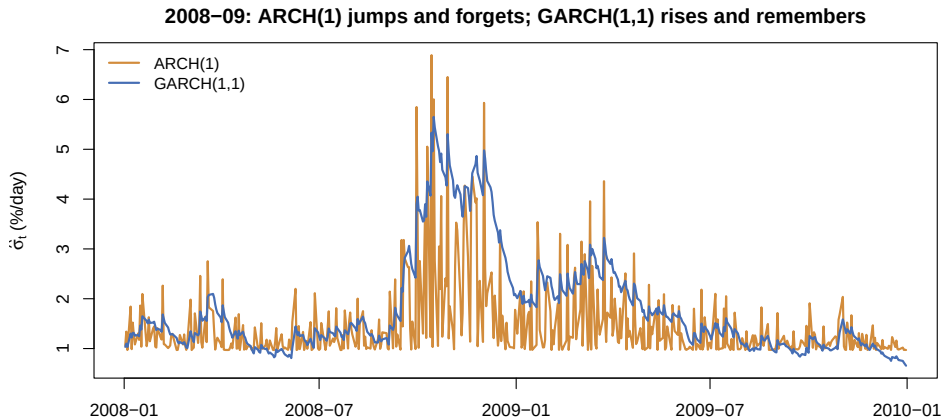
- Long-run $\hat{\sigma} = 1.14\%$ sits right by the sample s.d. 1.22% – a healthy sign.
- Textbook shape: **small $\hat{\alpha}$, large $\hat{\beta}$** – modest reaction, long memory ($H \approx 37$ days).
- Compare ARCH(1)'s $\hat{\alpha}_1 = 0.39$: GARCH splits that “reaction” into a gentle $\hat{\alpha} = 0.12$ *plus* inertia $\hat{\beta} = 0.86$.

The fitted volatility, laid over the returns



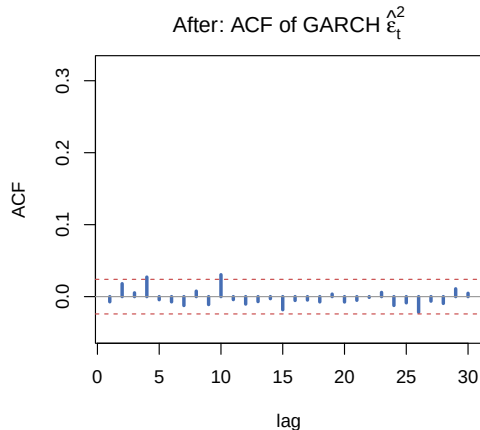
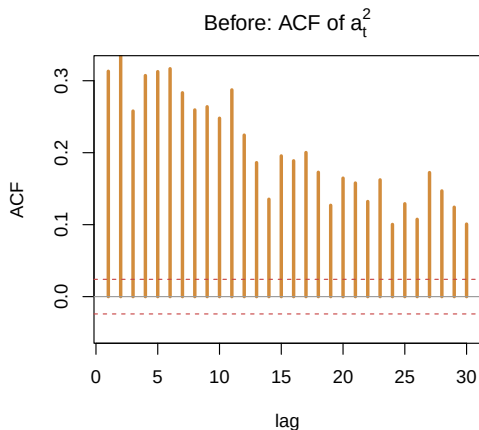
The $\pm 2\hat{\sigma}_t$ band **breathes** with the market – but, unlike ARCH(1), it rises and falls *smoothly*: the $\beta \sigma_{t-1}^2$ term carries yesterday's level forward, so the band glides into and out of crises instead of twitching day to day.

ARCH(1) jumps and forgets; GARCH(1,1) rises and remembers



The 2008–09 crisis, two conditional-volatility estimates. **ARCH(1) (orange)** spikes on each bad day and drops back the next – it has a one-day memory. **GARCH(1,1) (blue)** ratchets up and stays elevated for months, then eases down – the persistent memory (β) at work. GARCH's path is what actual volatility looks like.

Step 5 – did GARCH clean up the variance?



Left: the ACF of raw a_t^2 – large and persistent (the clustering we set out to model). **Right:** the ACF of the GARCH standardised squares $\hat{\varepsilon}_t^2$ – now inside the bands. *Ljung-Box(20)* on $\hat{\varepsilon}_t^2$: $p = 0.40$ (was ≈ 0 for ARCH). The volatility structure is gone.

Model comparison: GARCH(1,1) wins, and cheaply

ARCH(1) vs ARCH(5) vs GARCH(1,1) on daily S&P 500

Model	Params	AIC	BIC	Ljung-Box $\hat{\epsilon}_t^2$
ARCH(1)	2	20435	20449	≈ 0
ARCH(5)	6	18532	18573	9×10^{-10}
GARCH(1,1)	3	18276	18297	0.40

- **Lower AIC/BIC and fewer parameters:** GARCH(1,1) (3) beats ARCH(5) (6) on every count.
- It is the *only* model whose squared residuals pass Ljung-Box – the leftover ARCH is finally gone.
- **Remaining issue:** the standardised-residual kurtosis is still $4.75 > 3$. The tails need one more fix.

Student- t innovations

The tails, one more time: Student- t GARCH

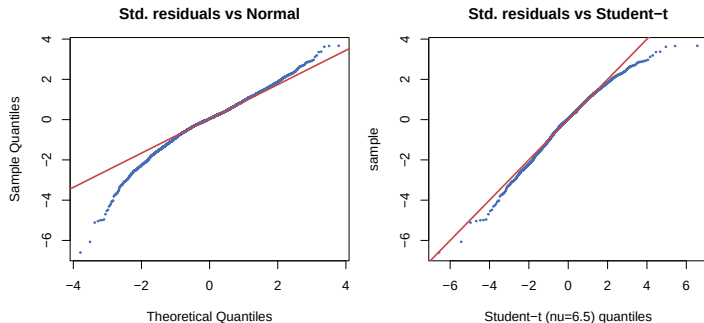
- GARCH now explains the *clustering* of extremes, but Gaussian innovations still under-count their *frequency*: residual kurtosis $4.75 \gg 3$.
- Same remedy as yesterday – swap $\varepsilon_t \sim \mathcal{N}(0, 1)$ for a **standardised Student- t_ν** and estimate ν jointly with ω, α, β by MLE.

Fitted GARCH(1, 1)- t on the S&P 500

$$\hat{\nu} \approx \mathbf{6.5}, \quad \hat{\alpha} + \hat{\beta} = 0.994.$$

- **Note** $\hat{\nu} \approx 6.5$, **far above ARCH's** $\hat{\nu} \approx 3.2$. GARCH's persistence already manufactures much of the kurtosis, so the innovations need *less* help – the two mechanisms share the work.
- Recall ν controls tail thickness and moment existence: $\nu > 4$ here, so even the fitted kurtosis is finite – a milder, more stable tail than ARCH implied.

The tails now fit



Left: GARCH standardised residuals still bend off the Normal line at the extremes. **Right:** against a Student- $t(\hat{\nu} \approx 6.5)$ they straighten out – location and clustering from GARCH, tail thickness from ν .

Forecasting volatility

Forecasting variance with GARCH(1, 1)

Stand at time t ; forecast the variance h steps ahead.

One step ahead – known

$$\sigma_t^2(1) = \mathbb{E}[a_{t+1}^2 \mid \mathcal{F}_t] = \omega + \alpha a_t^2 + \beta \sigma_t^2.$$

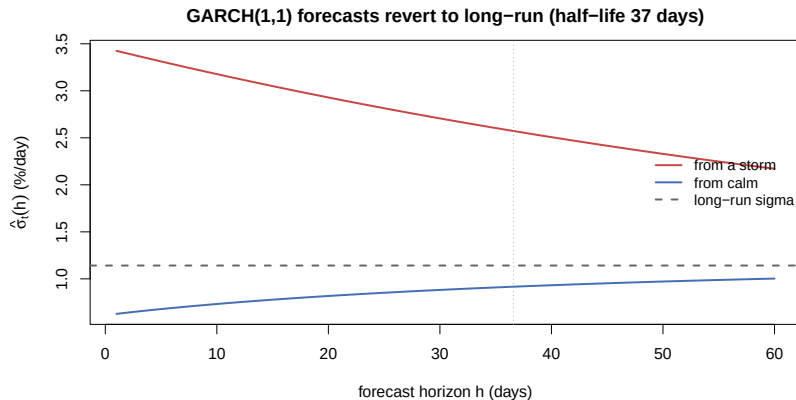
Both a_t and σ_t^2 are in hand, so tomorrow's variance carries *no* forecast error.

h steps ahead – recurse, using $\mathbb{E}[a_{t+h-1}^2 \mid \mathcal{F}_t] = \sigma_t^2(h-1)$

$$\sigma_t^2(h) = \omega + (\alpha + \beta) \sigma_t^2(h-1), \quad h = 2, 3, \dots$$

- Closed form: $\sigma_t^2(h) - \sigma^2 = (\alpha + \beta)^{h-1} (\sigma_t^2(1) - \sigma^2)$, with $\sigma^2 = \frac{\omega}{1 - \alpha - \beta}$.
- So the forecast **mean-reverts geometrically** to the long-run variance at rate $\alpha + \beta$ – half-way there in $H \approx 37$ days.

The forecast: from calm and from a storm



Two origins, same long-run anchor (dashed, 1.14%). From a **storm** the curve decays *down*; from **calm** it drifts *up*; both cross half their gap at $H \approx 37$ days (dotted). *ARCH's forecast reverted in days; GARCH's high persistence makes the whole term structure of volatility slope gently – the realistic picture.*

From variance forecast to a risk band

- As yesterday, a variance forecast becomes a **risk statement** through the innovation quantile:

$$r_{t+1} \in \hat{\mu}_{t+1} \pm q_{1-\alpha/2} \hat{\sigma}_t(1),$$

with $q = 1.96$ Gaussian, larger under Student- t .

- What GARCH adds over ARCH:** a *term structure* of risk. Because reversion is slow, the 10-day or 1-month VaR is *not* just $\sqrt{h} \times$ the daily one – it bends toward the long-run level at rate $(\alpha + \beta)^h$.

What GARCH buys the risk manager

A smooth, slowly-reverting volatility path and a coherent **multi-horizon** risk band – the daily breathing band of ARCH, now extended honestly to weeks and months ahead.

Does it travel? Cross-asset evidence

The same GARCH story on Apple and Microsoft

GARCH(1, 1) fitted to daily S&P 500, AAPL and MSFT

Asset	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha} + \hat{\beta}$	Half-life	Long-run $\hat{\sigma}$	LB $\hat{\varepsilon}_t^2$	$\hat{\nu}$
S&P 500	0.118	0.863	0.981	37 d	1.14%	0.40	6.5
AAPL	0.102	0.890	0.992	84 d	3.18%	0.99	4.7
MSFT	0.084	0.895	0.979	33 d	1.92%	0.99	4.5

- **Universal shape:** small $\hat{\alpha}$, large $\hat{\beta}$, persistence just under 1 everywhere – the “small reaction, long memory” signature is not an index quirk.
- **GARCH(1, 1) suffices for all three:** every $\hat{\varepsilon}_t^2$ Ljung–Box now passes ($p \geq 0.40$) – where ARCH needed many lags and still failed on the S&P.
- **Apple is both the most volatile and the most persistent** (long-run 3.2%/day, half-life ~ 4 months); single-name tails ($\hat{\nu} \approx 4.5$) are heavier than the index’s.

Bringing it together: a worked exercise

Exercise – a GARCH(1, 1) desk model

Setup

A desk models daily returns (in %) as a zero-mean GARCH(1, 1):

$$a_t = \sigma_t \varepsilon_t, \quad \varepsilon_t \stackrel{iid}{\sim} (0, 1), \quad \sigma_t^2 = 0.02 + 0.08 a_{t-1}^2 + 0.90 \sigma_{t-1}^2 \quad (\%^2).$$

Today closed with shock $a_t = +3\%$ and estimated volatility $\sigma_t = 1.2\%$.

Your tasks

- (a) Is the model **stationary**? State the condition and check it.
- (b) Find the **long-run** variance and daily volatility.
- (c) What is the **persistence**, and the volatility **half-life**?
- (d) Forecast **tomorrow's** variance $\sigma_t^2(1)$. Decompose it into baseline / news / memory.
- (e) Forecast **two days** ahead, $\sigma_t^2(2)$.
- (f) Give an approximate **95%** range for tomorrow's return (Gaussian ε).

Try (a)–(c) now; then we build (d)–(f) together.

Solution – parts (a)–(c)

(a) Stationarity. Need $\omega > 0$, $\alpha, \beta \geq 0$, $\alpha + \beta < 1$. Here $\alpha + \beta = 0.08 + 0.90 = 0.98 < 1$ ✓.

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(b) Long-run level.

$$\sigma^2 = \frac{\omega}{1 - \alpha - \beta} = \frac{0.02}{1 - 0.98} = \frac{0.02}{0.02} = 1.0 \text{ \%}^2 \implies \sigma = 1.0\% \text{ per day.}$$

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(c) Persistence and half-life.

$$\alpha + \beta = \mathbf{0.98}; \quad H = \frac{\ln 0.5}{\ln 0.98} = \frac{-0.693}{-0.0202} \approx \mathbf{34 \text{ trading days.}}$$

A volatility shock is half-gone only after about **seven weeks** – long memory, driven mostly by the big $\beta = 0.90$.

Solution – parts (d)–(f)

(d) One day ahead, using $a_t = 3$, $\sigma_t^2 = 1.2^2 = 1.44$:

$$\sigma_t^2(1) = \underbrace{0.02}_{\text{base}} + \underbrace{0.08(9)}_{\text{news} = 0.72} + \underbrace{0.90(1.44)}_{\text{memory} = 1.296} = \mathbf{2.04} \%^2 \Rightarrow \sigma_t(1) \approx 1.43\%.$$

Memory (1.30) dwarfs news (0.72) – the hallmark of high- β GARCH.

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$$\sigma_t^2(2) = 0.02 + 0.98(2.04) = \mathbf{2.02} \%^2 \Rightarrow \sigma \approx 1.42\% \quad (\text{barely falling – reversion is } \textit{slow}).$$

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(f) 95% range for tomorrow (Gaussian, mean 0):

$$0 \pm 1.96 \sigma_t(1) = \pm 1.96(1.43) \approx \pm \mathbf{2.8\%} \quad (\text{vs } \pm 1.96\% \text{ in normal times}).$$

What the exercise showed

One model, from three numbers and one day of data

- **Stationarity** ($\alpha + \beta < 1$) fixes a finite long-run variance $\omega/(1 - \alpha - \beta)$.
- **Persistence** $\alpha + \beta$ sets the half-life: how long a volatility shock lingers.
- **Tomorrow's risk is known today**, and *memory* usually dominates *news* in it.
- **Forecasts mean-revert slowly**, geometrically at rate $\alpha + \beta$ – a whole term structure.
- **The band breathes** and the multi-horizon VaR follows from $(\alpha + \beta)^h$.

Everything flowed from $\sigma_t^2 = \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2$ and today's (a_t, σ_t) .

Today in one slide

The problem

- ARCH captured clustering but needed *many* lags for equities' long memory.

The model: GARCH(1,1)

- $a_t = \sigma_t \varepsilon_t$, $\sigma_t^2 = \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2$.
- Constraints: $\omega > 0$, $\alpha, \beta \geq 0$, $\alpha + \beta < 1$.
- = ARCH(∞) with geometric weights $\alpha\beta^{j-1}$.

Properties

- Long-run var = $\omega/(1 - \alpha - \beta)$; persistence $\alpha + \beta$; half-life $\ln 0.5 / \ln(\alpha + \beta)$.
- Squares follow ARMA(1,1).

Practice

- Fit by MLE (recursion for σ_t^2); Student- t for tails.
- Forecast mean-reverts at rate $\alpha + \beta$: a term structure of risk.

The empirical thread

Daily S&P 500: $\hat{\alpha} \approx 0.12$, $\hat{\beta} \approx 0.86$, persistence 0.98, half-life 37 d, $\hat{\nu} \approx 6.5$. GARCH(1,1) (3 params) beats ARCH(5) (6) on AIC *and* passes every diagnostic.

Check your understanding

- 1 In $\sigma_t^2 = \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2$, what do α and β each control, and why is “small α , large β ” typical for equities?
- 2 Show GARCH(1, 1) is an ARCH(∞). What sets the rate at which old shocks fade?
- 3 Two assets have $\alpha + \beta = 0.90$ and 0.99. Which has the longer volatility half-life, and by roughly how much?
- 4 After a GARCH(1, 1) fit, $\hat{\varepsilon}_t^2$ is clean but the kurtosis is 5. What is left to fix, and how?
- 5 Why is the 20-day VaR from a GARCH *not* simply $\sqrt{20}$ times the 1-day VaR?

(Answers: reaction to news vs persistence of memory, markets digest news slowly; weights $\alpha\beta^{j-1}$, rate β ; the 0.99 asset – half-life 69 vs 6.6 days; fat tails $\rightarrow$ Student-t innovations; because volatility mean-reverts at rate $\alpha + \beta$, so multi-day risk bends toward the long-run level.)

Next time

From GARCH to asymmetric volatility

GARCH treats a +3% day and a -3% day identically – but markets don't: **bad news raises volatility more than good news** (the *leverage effect*). Next we let the model see the sign of the shock:

$$\sigma_t^2 = \omega + \alpha a_{t-1}^2 + \gamma a_{t-1}^2 \mathbf{1}_{\{a_{t-1} < 0\}} + \beta \sigma_{t-1}^2,$$

the **GJR-GARCH** (and its cousin, **EGARCH**). *That is Day 3.*

Questions?

Appendix – reproduce everything in R

All figures and numbers come from `Day2_GARCH_code.R` on the local `data/` folder.

```
r <- 100*diff(log(idx$SP500))    # S&P 500 daily log returns (%)
a <- r - mean(r)                 # mean-adjusted shocks

library(fGarch)
# GARCH(1,1), Gaussian and Student-t, by MLE
g  <- garchFit(~garch(1,1), data=a, cond.dist="norm", include.mean=FALSE)
gt <- garchFit(~garch(1,1), data=a, cond.dist="std",  include.mean=FALSE)

# persistence, long-run vol, half-life
ab <- alpha1 + beta1
lr <- sqrt(omega/(1-ab)); H <- log(0.5)/log(ab)

# forecast: sigma^2(h) = omega + (alpha+beta)*sigma^2(h-1)
```

Fitted (daily S&P 500): $\hat{\omega} = 0.0245$, $\hat{\alpha} = 0.118$, $\hat{\beta} = 0.863$, $\hat{\alpha} + \hat{\beta} = 0.981$, half-life 37 d, $\hat{\nu} \approx 6.5$; long-run $\hat{\sigma} = 1.14\%$.